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Catenyms

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Description

A catenym is a pair of words separated by a period such that the last letter of the first word is the same as the last letter of the second. For example, the following are catenyms:

```
dog.gopher
gopher.rat
rat.tiger
aloha.aloha
arachnid.dog
```

A compound catenym is a sequence of three or more words separated by periods such that each adjacent pair of words forms a catenym. For example,

```
aloha.aloha.arachnid.dog.gopher.rat.tiger
```

Given a dictionary of lower case words, you are to find a compound catenym that contains each of the words exactly once. The first line of standard input contains

 t

, the number of test cases. Each test case begins with 3 ≤

 n

≤ 1000 - the number of words in the dictionary.

 n

distinct dictionary words follow; each word is a string of between 1 and 20 lowercase letters on a line by itself. For each test case, output a line giving the lexicographically least compound catenym that contains each dictionary word exactly once. Output "****" if there is no solution.

Samples

input

Copy

```
2
6
aloha
arachnid
```

```
dog
gopher
rat
tiger
3
oak
maple
elm
```

output [Copy](#)

```
aloha.arachnid.dog.gopher.rat.tiger
***
```

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